

Removing the kernel zeros in Deddens's analytic-Toeplitz intertwining conjecture

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Abstract

Bourdon and Shapiro record as open the necessity half of Deddens's conjecture: if two analytic Toeplitz operators satisfy $XT_\varphi = T_\psi X$ for some nonzero bounded X , must $\overline{\psi(\mathbb{D})}$ lie in the point spectrum of T_φ^* ? We prove that it must. The usual reproducing-kernel argument gives an analytic eigenvector field except where X^* annihilates a kernel. At each such point, factor out the finite-order zero of this Hilbert-space-valued analytic field. The nonzero leading coefficient remains an eigenvector with the missing eigenvalue.

Status. This is a candidate new full proof of the necessity statement declared open in the introduction of [1]. A bounded search found no later resolution; no priority claim is made.

1 The source question

For $\varphi, \psi \in H^\infty(\mathbb{D})$, let T_φ and T_ψ denote multiplication by the symbols on the Hardy space H^2 . Deddens conjectured that

$$\overline{\psi(\mathbb{D})} \subseteq \sigma_p(T_\varphi^*) \quad (1)$$

is necessary and sufficient for the existence of a nonzero bounded operator X such that $XT_\varphi = T_\psi X$. Here the bar means pointwise complex conjugation, not closure. Cowen disproved sufficiency; Bourdon and Shapiro state that necessity remains open. Their kernel argument proves the desired point-spectrum assertion away from a discrete zero set and only a full-spectrum assertion at the remaining points.

Theorem 1. *Let $\varphi, \psi \in H^\infty(\mathbb{D})$. If a nonzero bounded operator $X : H^2 \rightarrow H^2$ satisfies*

$$XT_\varphi = T_\psi X,$$

then (1) holds. Thus the necessity half of Deddens's conjecture is true, without additional assumptions on either symbol.

2 Idea of the proof

The adjoint of an analytic Toeplitz operator has a canonical eigenvector at every point: the reproducing kernel K_a has eigenvalue $\overline{\psi(a)}$ for T_ψ^* . Applying X^* produces a T_φ^* -eigenvector unless $X^*K_a = 0$. These exceptional points are the zeros of a nonzero Hilbert-space-valued analytic function.

At an exceptional point, do not discard the eigenvector field. Divide it by the largest common power of the local coordinate. The intertwining identity survives off the zero, hence by continuity also at the zero; the value of the divided field there is nonzero and is precisely the missing eigenvector.

3 An analytic zero-removal lemma

Lemma 2. *Let A be a bounded operator on a complex Banach space E , let $F : \mathbb{D} \rightarrow E$ be analytic and not identically zero, and let $a : \mathbb{D} \rightarrow \mathbb{C}$ be analytic. If*

$$AF(\zeta) = a(\zeta)F(\zeta) \quad (\zeta \in \mathbb{D}), \quad (2)$$

then $a(\mathbb{D}) \subseteq \sigma_p(A)$.

Proof. Fix $\zeta_0 \in \mathbb{D}$. If $F(\zeta_0) \neq 0$, equation (2) gives the assertion. Otherwise, the Taylor expansion of the nonzero Banach-valued analytic function F has a first nonzero coefficient at some finite order $m \geq 1$. Thus locally

$$F(\zeta) = (\zeta - \zeta_0)^m G(\zeta),$$

where G is analytic and $G(\zeta_0) \neq 0$. Cancelling the scalar factor in (2) for $\zeta \neq \zeta_0$ gives $AG(\zeta) = a(\zeta)G(\zeta)$. Continuity at ζ_0 yields

$$AG(\zeta_0) = a(\zeta_0)G(\zeta_0), \quad G(\zeta_0) \neq 0.$$

Hence $a(\zeta_0)$ is an eigenvalue. Since ζ_0 was arbitrary, the lemma follows. $\square$

4 Resolution of the conjectured necessity

Proof of Theorem 1. For $\zeta \in \mathbb{D}$, define the Hardy-space vector

$$k_\zeta(w) = \frac{1}{1 - \zeta w} \quad (w \in \mathbb{D}).$$

The map $\zeta \mapsto k_\zeta$ is H^2 -valued analytic; in coefficient coordinates it is $(1, \zeta, \zeta^2, \dots)$. Put

$$F(\zeta) = X^* k_\zeta, \quad \psi^\sharp(\zeta) = \overline{\psi(\bar{\zeta})}.$$

Both functions are analytic. Moreover, F is not identically zero. Indeed, $k_\zeta = K_{\bar{\zeta}}$, and the span of all Hardy reproducing kernels is dense; if X^* killed every k_ζ , then $X^* = 0$ and hence $X = 0$.

Taking adjoints in the intertwining identity gives $T_\varphi^* X^* = X^* T_\psi^*$. Since

$$T_\psi^* k_\zeta = \psi^\sharp(\zeta) k_\zeta,$$

we obtain the analytic eigenfield identity

$$T_\varphi^* F(\zeta) = \psi^\sharp(\zeta) F(\zeta) \quad (\zeta \in \mathbb{D}).$$

Lemma 2 therefore implies $\psi^\sharp(\mathbb{D}) \subseteq \sigma_p(T_\varphi^*)$. Finally, for each $a \in \mathbb{D}$ take $\zeta = \bar{a}$ to get

$$\psi^\sharp(\bar{a}) = \overline{\psi(a)} \in \sigma_p(T_\varphi^*).$$

This is exactly (1). $\square$

Verification notes

The conjugate parameter $\zeta = \bar{a}$ is used solely to make the kernel field analytic. The zero-removal step is valid for arbitrary Banach-valued analytic functions: nonidentity to zero guarantees a first nonzero Taylor coefficient at every zero. The proof establishes point spectrum, not merely spectrum, because the divided field is nonzero at every formerly exceptional point. The source question page and a separate audit accompany this packet.

References

- [1] P. S. Bourdon and J. H. Shapiro, *Intertwining relations and extended eigenvalues for analytic Toeplitz operators*, Illinois J. Math. 52 (2008), 1007–1030; arXiv:0801.1972.